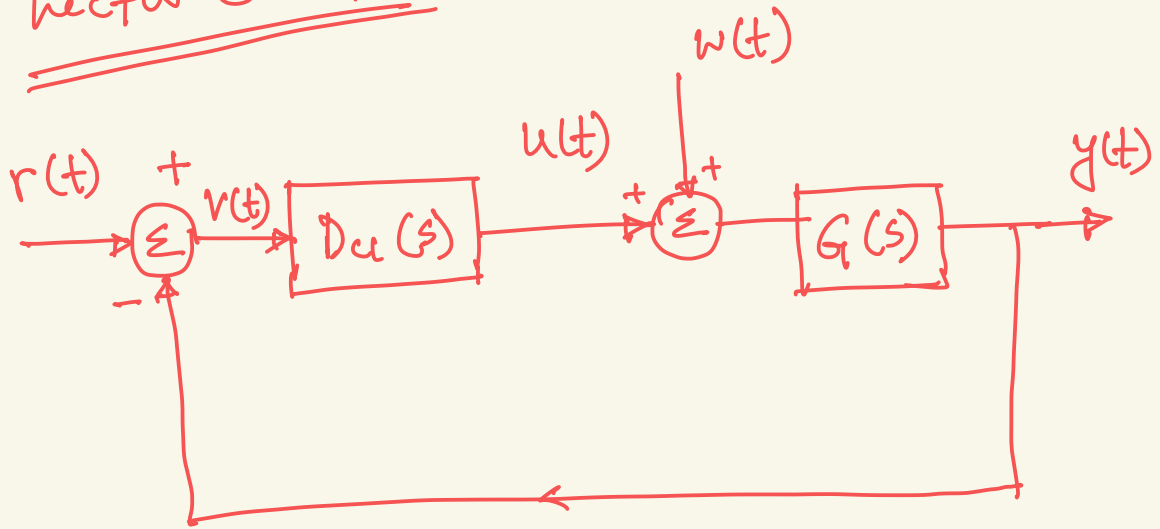




Lecture 24



Consider the unity feedback system drawn above.

$r(t)$ = reference signal

$v(t) = r(t) - y(t)$ = error = $e(t)$

$y(t)$ = output signal

$G(s)$ = system transfer function

$D_c(s)$ = closed loop transfer function

$w(t)$ = noise or disturbance

$$v(t) = r(t) - y(t)$$

$$\Rightarrow V(s) = R(s) - Y(s)$$

$$U(s) = D_u(s) V(s)$$

$$\Rightarrow U(s) = D_u(s) [R(s) - Y(s)]$$

$$Y(s) = G(s) [U(s) + W(s)]$$

$$= G(s) [D_u(s) [R(s) - Y(s)] + W(s)]$$

$$Y(s) = G(s) D_u(s) R(s) - Y(s) D_u(s) G(s) + G(s) W(s)$$

$$Y(s) + Y(s) D_u(s) G(s) = G(s) D_u(s) R(s) + G(s) W(s)$$

$$\Rightarrow Y(s) [1 + D_u(s) G(s)] = G(s) D_u(s) R(s) + G(s) W(s)$$

$$\Rightarrow Y(s) = \frac{G(s) D_u(s)}{1 + D_u(s) G(s)} R(s) + \frac{G(s) W(s)}{1 + D_u(s) G(s)}$$

transfer function
(closed loop)

sensitivity function

closed loop transfer function

$$T(s) = \frac{G(s) D_u(s)}{1 + D_u(s) G(s)}$$

